

Jong Yeon Lee

Assistant Professor of Physics

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Experience

University of Illinois Urbana-Champaign , Assistant Professor, Department of Physics	Urbana, IL
• Anthony Leggett Institute of Condensed Matter Theory	2024 – present
• Illinois Quantum Information Science and Technology Center (IQUIST)	2 years
Korea Institute for Advanced Study , Quantum Universe Scholar	Seoul, Korea
	2025 – present
	1 year
University of California, Berkeley , Research Scholar	Berkeley, CA
• Host: Ehud Altman	2023 – 2024
	1 year
Kavli Institute for Theoretical Physics , Gordon and Betty Moore Postdoctoral Fellow	Santa Barbara, CA
• Hosts: Matthew Fisher and Cenke Xu	2020 – 2023
	3 years

Education

PhD Harvard University , Physics	Cambridge, MA
• Advisor: Ashvin Vishwanath; thesis committee: Subir Sachdev and Philip Kim	2015 – 2020
• Thesis: <i>Fractionalization, Emergent Gauge Dynamics, and Topology in Quantum Matter</i>	
BS/MS California Institute of Technology , Physics and Mathematics	Pasadena, CA
• Academic advisor: John Preskill	2011 – 2015
• Rank 2 of 239	

Awards and Honors

2026: Microsoft Quantum Pioneer Award
2026: KSEA Young Investigator Grant Award
2025–2028: IBM-IIDAI Research Grant
2022: Outstanding Young Researcher Award (for outstanding achievements by junior Korean physicists)
2020–2023: Gordon and Betty Moore Fellowship, Kavli Institute of Theoretical Physics
2015: Richard P. Feynman Prize in Theoretical Physics, Caltech (awarded annually to a senior who has demonstrated excellence in theoretical physics)
2011–2015: Korean Presidential Scholarship (granted annually to exceptional students from Korea to study abroad)
2011: Physics Research Tutorial Fellowship, Caltech (granted annually to a small group of freshmen)
2007: International Astronomy Olympiad, Gold Medal (top place)

Service

- Organizer, [Quantum Information and Dynamics 2026: qLDPC Theory and Application](#), KIAS, Seoul, July 6–9, 2026
- Organizer, [Quantum Information and Dynamics 2025: Non-equilibrium Physics](#), Seoul, June 23–27, 2025
- Organizer, IQUIST seminar, 2025–2026
- Organizer, Institute of Condensed Matter Theory seminar, 2025–2026
- Faculty hiring committee for Quantum Information Theory, 2025–26
- Organizer, UIUC–UChicago joint workshop, 2025

- Referee for *Science*, *Nature*, *Physical Review Letters*, *Physical Review X*, *PRX Quantum*, *Physical Review A*, *Physical Review B*, and *Physical Review E*
- Editorial Advisory Board member, APS Open Science

Teaching

- Physics 213 / 214 — University Physics: Thermal Physics and Quantum Physics, University of Illinois Urbana-Champaign, Fall 2026
- Physics 504 — Statistical Physics, University of Illinois Urbana-Champaign, Spring 2026
- Physics 598 — Many-Body Physics of Information, University of Illinois Urbana-Champaign, Spring 2025

Invited Talks

- Microsoft Quantum Pioneer Forum, Aug 14, 2026
- Yukawa Institute for Theoretical Physics, [Emergent Phenomena in Quantum Matter: From Symmetry to Dynamics](#), Jun 30, 2026
- Yukawa Institute for Theoretical Physics, [Thermalization and Hydrodynamics](#), Jun 4, 2026
- Simons Center for Geometry and Physics, [Bridging Classic and Contemporary Perspectives on Open Quantum Systems](#), May 6, 2026
- KITP Conference, [Frontiers of Programmable Quantum Dynamics](#), Oct 30, 2025
- Condensed Matter Seminar, California Institute of Technology, Oct 27, 2025
- [Ultra Quantum Matter Simons Workshop](#), Chicago, Jun 2–3, 2025
- Q-farm Seminar, Stanford University, Apr 12, 2025
- Kadanoff Seminar, University of Chicago, Feb 24, 2025
- Condensed Matter Theory Seminar, Texas A&M University, Feb 14, 2025
- Condensed Matter Seminar, University of Texas at Austin, Feb 13, 2025
- [Focus Week on Non-equilibrium Quantum Dynamics](#), Kavli IPMU, University of Tokyo, Oct 3, 2024
- Simons Center Program, [Non-Hermitian Topology, Geometry and Symmetry across Physical Platforms](#), Sep 20, 2024
- [KIAS Summer School in Quantum Information](#), Aug 12, 2024
- [Chez Pierre Seminar](#), MIT, May 22, 2024
- KIAS-IBS-PCS Workshop on Correlation and Topology in Quantum Matter, Dec 18–21, 2023
- New Horizons in Condensed Matter Physics 2023, Nov 3–5, 2023
- US-Korea Conference 2023 on Science, Technology, and Entrepreneurship, Aug 5, 2023
- Quantum Colloquium, Korea University, May 17, 2023
- Special Seminar, Korea Institute for Advanced Study, May 10, 2023
- Special Seminar, Pohang University of Science and Technology, May 9, 2023
- Special Seminar, Seoul National University, May 4, 2023
- Special Seminar, Binghamton University, Mar 3, 2023
- Quantum Information Seminar, Purdue University, Feb 23, 2023
- Special Seminar, University of Pittsburgh, Feb 21, 2023
- Special Condensed Matter Physics Seminar, Northwestern University, Feb 15, 2023
- Physics and Astronomy Colloquium, Rice University, Feb 13, 2023
- Institute of Condensed Matter Physics Seminar, UIUC, Feb 9, 2023
- IQUIST Seminar, UIUC, Feb 7, 2023
- KAIX Future of Physics Workshop, Dec 27, 2022
- KITP Program: Noisy Intermediate-Scale Quantum Systems, Sep 22, 2022
- Condensed Matter Seminar, Caltech, Sep 7, 2022
- Gordon Research Seminar: Correlated Electron Systems, Jun 26, 2022
- AKPA-KPS Symposium, May 15, 2022
- Physics Colloquium, Carnegie Mellon University, Feb 7, 2022
- Correlated Electrons Virtual International Seminars (CEVIS), May 27, 2021

- Center of Mathematical Sciences and Applications, Apr 1, 2020
- Condensed Matter Seminar, Institute of Solid State Physics, Japan, Jan 17, 2020
- Microsoft Quantum Research Institute, Oct 22, 2019
- The 3rd EPIQS-TMS Alliance Workshop on Topological Phenomena in Quantum Materials (TPQM2019), Oct 21, 2019
- The 5th Conference on Condensed Matter Physics (CCMP-2019), China, invited session, Jun 29, 2019
- Invited Seminar, Perimeter Institute for Theoretical Physics, Apr 16, 2019
- Condensed Matter Physics Seminar, Korea Institute for Advanced Study, Jan 11, 2019
- Condensed Matter Physics Seminar, Korea Advanced Institute of Science and Technology (KAIST), Jan 7, 2019
- Condensed Matter Seminar, Institute of Solid State Physics, Japan, Dec 18, 2018
- Condensed Matter Seminar, MIT, Sep 17, 2018